

EXACT STATE PARTITION, ZETA FUNCTION, AND CRITICAL WINDOW FOR MAJORITY DYNAMICS ON COCKTAIL-PARTY GRAPHS

ANONYMOUS

ABSTRACT. Let $CP_n = K_{2n} \setminus M$ be the cocktail-party graph obtained by deleting a perfect matching, and apply synchronous strict majority with inertia at a tie. We determine the image of every one of its 4^n Boolean states after one step. The recurrent core consists of $2 + 2^n$ fixed states and

$$\frac{1}{2} \left\{ \binom{2n}{n} - 2^n \right\}$$

genuine two-cycles; every other state reaches consensus in one step. This gives exact consensus-basin sizes, every iterate fixed-point count, the Artin–Mazur zeta function, and the complete symbolic natural extension. For i.i.d. Bernoulli initial data we also obtain exact outcome probabilities. In the critical window $p = 1/2 + a/\sqrt{n}$, the probability of one-consensus tends to $\Phi(2\sqrt{2}a)$, while the genuine two-cycle probability is asymptotic to $e^{-4a^2}/\sqrt{\pi n}$. An exhaustive implementation of the literal local rule checks all states through $n = 8$ and the first twelve iterate counts.

1. INTRODUCTION

Synchronous Boolean threshold networks with symmetric interaction have no cycles longer than two, a foundational fact due to Goles and Olivos [3]. For majority dynamics with the convention that a vertex retains its state at a tie, Ginosar and Holzman study the same local rule on infinite graphs [2]. Quantitative work on finite majority processes has emphasized convergence time and graph modules [4]. Those general results tell us the maximum possible period, but do not by themselves enumerate the recurrent states or their basins on a specified graph family.

This note performs that enumeration for the cocktail-party graph. Its vertices come in n nonadjacent mate pairs, while every two vertices from different pairs are adjacent. A global imbalance forces consensus in one step. At exact balance, each mate pair evolves independently: a monochrome pair flips and a mixed pair freezes. This elementary dichotomy yields a complete state partition and, from it, several dynamical quantities that are usually studied separately.

Our scope is intentionally exact and family-specific. We use the general period-two theorem as prior art, and do not treat irreversible conversion processes, whose multipartite theory has a different update rule [1]. The residual contribution is the joint package of state counts, basin counts, zeta function, natural extension, and Bernoulli critical-window asymptotics for the symmetric synchronous rule on CP_n .

2. THE LOCAL RULE

Fix $n \geq 1$. Write the vertices of CP_n as

$$V_n = \{1, \bar{1}, \dots, n, \bar{n}\},$$

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where i and $\bar{i}$ are mates and hence are the only distinct nonadjacent pair involving either vertex. A state is $x = (x_v)_{v \in V_n} \in \{0, 1\}^{V_n}$. Let

$$S(x) = \sum_{v \in V_n} x_v.$$

Every vertex has degree $2n - 2$. The synchronous majority map $F_n : \{0, 1\}^{V_n} \rightarrow \{0, 1\}^{V_n}$ is

$$(F_n x)_v = \begin{cases} 1, & \sum_{w \sim v} x_w > n - 1, \\ 0, & \sum_{w \sim v} x_w < n - 1, \\ x_v, & \sum_{w \sim v} x_w = n - 1. \end{cases}$$

Thus a tie is inertial. This convention also makes F_1 the identity on the four states of the edgeless graph CP_1 .

Lemma 2.1 (one-step trichotomy). *For every $x \in \{0, 1\}^{V_n}$:*

- (1) *if $S(x) < n$, then $F_n x = 0^{V_n}$;*
- (2) *if $S(x) > n$, then $F_n x = 1^{V_n}$;*
- (3) *if $S(x) = n$, then on each mate pair $\{i, \bar{i}\}$ the map fixes the patterns 01 and 10 and interchanges 00 with 11.*

Proof. The number of one-neighbors seen by v is

$$N_v(x) = S(x) - x_v - x_{\bar{v}}.$$

If $S(x) < n$, then $N_v(x) \leq n - 1$. Equality can occur only when $S(x) = n - 1$ and the mate pair of v is 00, so inertia still leaves v equal to zero. Otherwise $N_v(x) < n - 1$ and v becomes zero. This proves the first assertion; complementation proves the second.

Suppose $S(x) = n$. A 00 pair has $N_v(x) = n$ at both vertices and becomes 11. A 11 pair has $N_v(x) = n - 2$ and becomes 00. A mixed pair has $N_v(x) = n - 1$ at both vertices, so both coordinates retain their values. $\square$

3. COMPLETE FINITE DYNAMICS

Call a state *pairwise mixed* if each mate pair is either 01 or 10. There are exactly 2^n such states, all of weight n .

Theorem 3.1 (state partition). *For $n \geq 1$, the recurrent core of F_n has:*

$$\begin{aligned} \#\{\text{fixed states}\} &= 2 + 2^n, \\ \#\{\text{genuine two-cycles}\} &= \frac{1}{2} \left\{ \binom{2n}{n} - 2^n \right\}. \end{aligned}$$

The fixed states are the two consensus states and the 2^n pairwise-mixed states. Each full consensus basin has cardinality

$$B_n = \frac{4^n - \binom{2n}{n}}{2}.$$

Every nonrecurrent state enters the recurrent core after exactly one step, and $F_n^3 = F_n$.

Proof. By lemma 2.1, all states of weight below n enter zero-consensus and all states of weight above n enter one-consensus. Complementation is a bijection between these two sets, while the middle layer has size $\binom{2n}{n}$. Hence each consensus basin, including its consensus state, has size

$$\sum_{j=0}^{n-1} \binom{2n}{j} = \frac{4^n - \binom{2n}{n}}{2}.$$

On the middle layer, pairwise-mixed states are fixed. Every other state has at least one monochrome mate pair; all monochrome pairs flip and all mixed pairs freeze. A second step therefore returns the original state, but the first step does not. Thus the remaining $\binom{2n}{n} - 2^n$ middle-layer states form genuine two-cycles. The same description shows that every image is recurrent and proves $F_n^3 = F_n$. $\square$

The formula is more informative than the general statement that symmetric threshold networks have period at most two: it records the complete functional graph up to the individual leaves attached to the two consensus states.

4. ITERATE COUNTS, ZETA FUNCTION, AND NATURAL EXTENSION

For a map on a finite set, define its Artin–Mazur zeta function by

$$\zeta_{F_n}(z) = \exp \left(\sum_{k \geq 1} \frac{|\text{Fix}(F_n^k)|}{k} z^k \right).$$

Corollary 4.1 (iterate counts and zeta). *For every $k \geq 1$,*

$$|\text{Fix}(F_n^k)| = \begin{cases} 2 + 2^n, & k \text{ odd}, \\ 2 + \binom{2n}{n}, & k \text{ even}. \end{cases}$$

Consequently

$$\zeta_{F_n}(z) = (1 - z)^{-(2+2^n)} (1 - z^2)^{-\frac{1}{2} \{ \binom{2n}{n} - 2^n \}}.$$

Proof. An odd iterate fixes exactly the fixed states. An even iterate additionally fixes every point on a two-cycle. The Euler product of a finite dynamical system contributes $(1 - z^\ell)^{-1}$ for each cycle of length ℓ , giving the displayed formula. $\square$

There is also a canonical invertible symbolic system associated with the noninvertible map. Its natural extension is

$$\widehat{X}_n = \{(x_t)_{t \in \mathbb{Z}} : F_n(x_t) = x_{t+1} \text{ for every } t\}, \quad \widehat{\sigma}((x_t)_t) = (x_{t+1})_t.$$

Corollary 4.2 (symbolic natural extension). *The projection $(x_t)_t \mapsto x_0$ conjugates $(\widehat{X}_n, \widehat{\sigma})$ to the restriction of F_n to its recurrent core. Hence $\widehat{X}_n$ is the disjoint union of $2 + 2^n$ fixed orbits and $\frac{1}{2} \{ \binom{2n}{n} - 2^n \}$ orbits of length two, and its zeta function is the one in [corollary 4.1](#).*

Proof. In a finite functional graph, a point has prehistories of every length that fit into a bi-infinite orbit if and only if it lies on a directed cycle. Theorem 3.1 identifies all such points. On this recurrent core F_n is bijective, so the zeroth coordinate uniquely determines the entire bi-infinite orbit. $\square$

5. BERNOULLI INITIAL DATA AND THE CRITICAL WINDOW

Let the $2n$ initial coordinates be independent Bernoulli variables of parameter $p \in [0, 1]$. Write $H_n \sim \text{Bin}(2n, p)$.

Theorem 5.1 (exact outcome law). *Under Bernoulli(p) initial data,*

$$\begin{aligned} \mathbb{P}(\text{one-consensus}) &= \mathbb{P}(H_n > n), \\ \mathbb{P}(\text{zero-consensus}) &= \mathbb{P}(H_n < n), \\ \mathbb{P}(\text{nonconsensus fixed state}) &= [2p(1 - p)]^n, \\ \mathbb{P}(\text{genuine two-cycle}) &= \left\{ \binom{2n}{n} - 2^n \right\} [p(1 - p)]^n. \end{aligned}$$

These four events partition the probability space.

Proof. The first two identities are [lemma 2.1](#). Every pairwise-mixed state contains exactly n ones and has probability $p^n(1 - p)^n$; there are 2^n such states. Every other balanced state lies on a genuine two-cycle, and there are $\binom{2n}{n} - 2^n$ of them. $\square$

At $p = 1/2$, the genuine two-cycle probability is

$$\frac{\binom{2n}{n} - 2^n}{4^n} = \frac{1}{\sqrt{\pi n}} (1 + O(n^{-1})) - 2^{-n}.$$

The next result resolves the whole $n^{-1/2}$ bias window.

Corollary 5.2 (critical window). *Fix $a \in \mathbb{R}$ and put $p_n = 1/2 + a/\sqrt{n}$, for all sufficiently large n . Then*

$$\begin{aligned}\mathbb{P}_{p_n}(\text{one-consensus}) &\longrightarrow \Phi(2\sqrt{2}a), \\ \mathbb{P}_{p_n}(\text{zero-consensus}) &\longrightarrow \Phi(-2\sqrt{2}a), \\ \mathbb{P}_{p_n}(\text{genuine two-cycle}) &\sim \frac{e^{-4a^2}}{\sqrt{\pi n}}.\end{aligned}$$

The nonconsensus fixed-state probability is exponentially small.

Proof. For $H_n \sim \text{Bin}(2n, p_n)$,

$$\frac{n - 2np_n}{\sqrt{2np_n(1 - p_n)}} \longrightarrow -2\sqrt{2}a.$$

The central limit theorem gives the first two limits; the atom at n is $O(n^{-1/2})$ and does not affect them. Stirling's formula and

$$[p_n(1 - p_n)]^n = 4^{-n}(1 - 4a^2/n)^n \sim 4^{-n}e^{-4a^2}$$

give the two-cycle asymptotic. Finally, $[2p_n(1 - p_n)]^n = 2^{-n}(1 - 4a^2/n)^n$. $\square$

6. DETERMINISTIC CONTROL AND CLAIM BOUNDARY

The companion script constructs the deleted-matching neighbor list and evaluates the literal local majority rule; it does not use the trichotomy as an update shortcut. It exhausts all 4^n states for $1 \leq n \leq 8$, checks $F_n^3 = F_n$, the state and basin formulas, and $|\text{Fix}(F_n^k)|$ for $1 \leq k \leq 12$. This is a finite regression control, not a substitute for the proofs.

General period-two behavior is explicitly attributed to Goles and Olivos [3], and the tie convention to the established majority literature [2]. A bounded search through 27 August 2026 found no exact collision with the combined cocktail-party formulas in [theorems 3.1](#) and [5.1](#) and [corollaries 4.1](#), [4.2](#) and [5.2](#); this is a search record, not a worldwide priority claim. Public release and venue selection remain outside the scope of this internal draft.

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